

1. String Slicing with Examples:

String slicing allows extracting parts of a string using indices.

Example:

```
s = "Python"
```

```
print(s[0:2]) # Py
```

```
print(s[-3:]) # hon
```

2. Key Features of Lists in Python:

Lists are ordered, mutable, allow duplicates, and can store mixed data types.

Example: `lst = [1, "apple", 3.5]`

3. Access, Modify, and Delete Elements in a List:

Access: `lst[0]`

Modify: `lst[1] = "banana"`

Delete: `del lst[2]`

4. Difference Between Tuples and Lists:

Lists are mutable, tuples are immutable.

Example:

```
lst = [1,2,3]
```

```
tpl = (1,2,3)
```

5. Key Features of Sets:

Sets are unordered, do not allow duplicates, and support set operations.

Example: `s = {1,2,3}`

6. Use Cases of Tuples and Sets:

Tuples: used for fixed data.

Sets: used for removing duplicates and membership testing.

7. Add, Modify, Delete in Dictionary:

Add: `d["a"] = 1`

Modify: `d["a"] = 2`

Delete: `del d["a"]`

8. Importance of Immutable Dictionary Keys:

Dictionary keys must be immutable so they can be hashed.

Example: valid key: `(1,2)`, invalid key: `[1,2]`